PHYSIOLOGY OF THE MOTOR SYSTEM

Control of gaze

Lecture 03

A · The eyes

	Somatic	Eye
Effector	~200 joints, ~600 muscles	one sphere, three muscle pairs
Degrees of freedom	more than any task needs	yoked: direction + depth
Load	varies with what you hold	the orbit's own, predictable
Motor unit	up to ~1,800 fibres	~5 fibres
Feedback	muscle sense, touch, vision	vision; a different error for each

Kandel 6e, Chs. 30, 31 and 35 · teaching contrast, row by row; not a wiring diagram

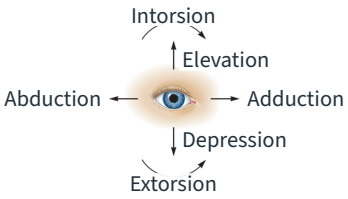
> Control of eye movements must be different from control of arm movements.

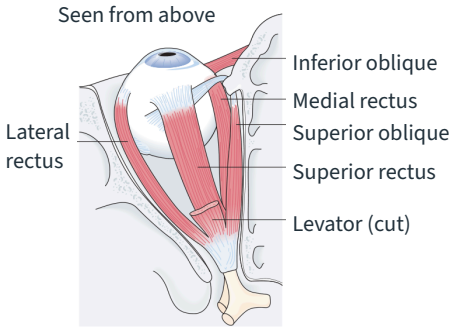
LECTURE NOTE

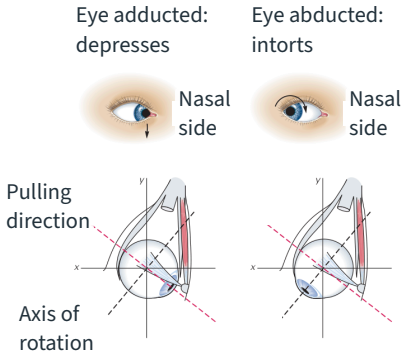
The eye differs from the arm in almost every mechanical respect: one sphere instead of many joints, degrees of freedom removed by yoking, tiny motor units, a predictable load, and correction by vision rather than by muscle sense. Its control must differ to match. This section takes the eye apart as an effector.

YOUR NOTES

The eye is controlled by 3 pairs of muscles







Kandel 6e, Figs. 35–1A and B2, p. 861 and 35–2, p. 862 · extraocular muscles; labels re-set from the figures

- Six muscles form three agonist–antagonist pairs, one for each rotation axis.
- The superior oblique:
  - Depresses adducted eye
  - Intorts abducted eye
- So what a vertical muscle does depends on where the eye already points.

LECTURE NOTE

Six muscles in three pairs rotate the globe horizontally, vertically and about the line of sight. The superior oblique depresses the adducted eye but intorts the abducted eye, so the brain cannot map one movement onto one muscle; the command must depend on current eye position.

YOUR NOTES

Wired yoking makes the two eyes move as one

Conjugate gaze to the right

Right eye	Left eye
lateral rectus	medial rectus
superior rectus (up)	inferior oblique (up)
inferior rectus (down)	superior oblique (down)

Degrees of freedom

- Yoking leaves **direction** and **depth** as the 3 main degrees of freedom
  - Direction: horizontal and vertical
  - Depth: vergence

- Each muscle has a yoke partner in the other eye, and the two are driven together.
- Extra degrees of freedom:
  - Eyes: removed by wiring, not control
  - Arm: more than any task needs; the controller must choose

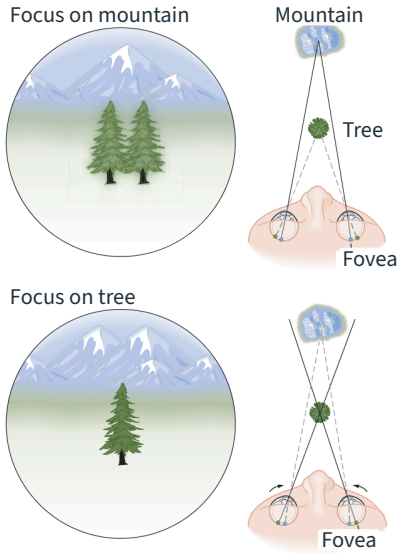
Kandel 6e, Ch. 35, pp. 862–863; Ch. 30, p. 714 · yoke pairs from the text; Table 35–1 gives the vertical muscles' actions

LECTURE NOTE

Each extraocular muscle is yoked to a partner in the other eye, so the eyes move as one. Yoking leaves three main degrees of freedom: horizontal and vertical direction, and depth through vergence. The arm has more degrees of freedom than any task needs and the controller must choose; for the eyes, wiring removes them.

YOUR NOTES

Vergence movements break the yoke



- Only the horizontal recti work

- Far to near, the eyes turn in opposite directions.
- It supplies depth, the freedom the yoke leaves.

Kandel 6e, Fig. 35–16, p. 879 · schematic after F. A. Miles; labels re-set from the figure

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LECTURE NOTE

Vergence turns the eyes toward or away from each other to put a near or far object on both foveae. It is the one disconjugate eye movement and uses only the horizontal recti. It adds the depth dimension to gaze direction.

YOUR NOTES

Eye muscles have tiny motor units

Innervation number

≈ 5
fibres per motor neuron, eye muscle

≈ 1,800
fibres per motor neuron, leg muscle

The load

- Only the orbit's own viscosity and elasticity
  - No object to lift, no ground to push against
- The eye "tends not to be substantially perturbed by external events"

- Eye force is finely graded; fibres per motor neuron:
  - Eye: about 5
  - Leg: about 1,800
- With a predictable load, the force for a movement can be known in advance.
- So commands can be issued without waiting for feedback.

Kandel 6e, Table 31–1, p. 739; Ch. 30, p. 717

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LECTURE NOTE

Eye muscles have about five fibres per motor neuron against about 1,800 in a leg muscle, and the only load is the orbit's own viscosity and elasticity. Because the eye is rarely perturbed from outside, the command for a given movement can be computed in advance and issued without waiting for feedback.

YOUR NOTES

Section A summary: the eye is a simple, predictable effector

	Somatic	Eye
Effector	~200 joints, ~600 muscles	one sphere, three muscle pairs
Degrees of freedom	more than any task needs	yoked: direction + depth
Load	varies with what you hold	the orbit's own, predictable
Motor unit	up to ~1,800 fibres	~5 fibres
Feedback	muscle sense, touch, vision	vision; a different error for each

TAKE HOME

- Few degrees of freedom: yoked; vergence breaks the yoke.
- Tiny motor units and a predictable load: forces can be planned in advance.

Kandel 6e, Chs. 30, 31 and 35 · the section A table, returning

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LECTURE NOTE

The section opened with this table and closes with it. The eye is a simple effector: one sphere with three muscle pairs, degrees of freedom removed by yoking except for vergence, tiny motor units, and a load that is the orbit's own and predictable, so commands can be computed in advance.

YOUR NOTES

B · Gaze and reaching

	Somatic	Eye
Final common path	motor neurons and muscles	oculomotor nuclei and muscles
Systems	posture, locomotion, reach, grasp	six: one shift, five holds
Overlap	shared spinal interneurons	shared brainstem nuclei, FEF
Control	feedforward and feedback	feedforward and feedback
Response latency	25 ms (reflex) to 200 ms (visual)	15 ms (VOR) to 200 ms (saccade)

Kandel 6e, Chs. 30, 33 and 35; Kandel 5th ed., Ch. 40; response latencies: instructor, 2026-11-08 · teaching comparison, row by row; not a wiring diagram

> Control of gaze is organized like control of the body.

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LECTURE NOTE

The eye is a different effector, but its control is organized as L01 described for the body: many systems converge on one final common path, the systems overlap in part, and each mixes feedforward and feedback control on feedback channels with different delays. This section sets the two side by side.

YOUR NOTES

One set of muscles serves many kinds of movement

Body

- Functions: eye movement, reach and grasp, posture, locomotion, breathing, speech
  - Served by overlapping groups of muscles
- Breathing muscles, in every control mode:
  - Voluntary: deep breath before a dive
  - Rhythmic: regular breathing
  - Reflexive: cough

Eye

- Fixating, shifting, following, converging, stabilizing
- The oculomotor nuclei are the final common targets of every type of eye movement

- Body: one set of muscles serves many functions and all three control modes.
- Eye: every type of movement ends on the same motor neurons.
- In both, many systems converge on one final common path: the motor neurons and their muscles.

Kandel 6e, Ch. 30, p. 715; Ch. 35, p. 866

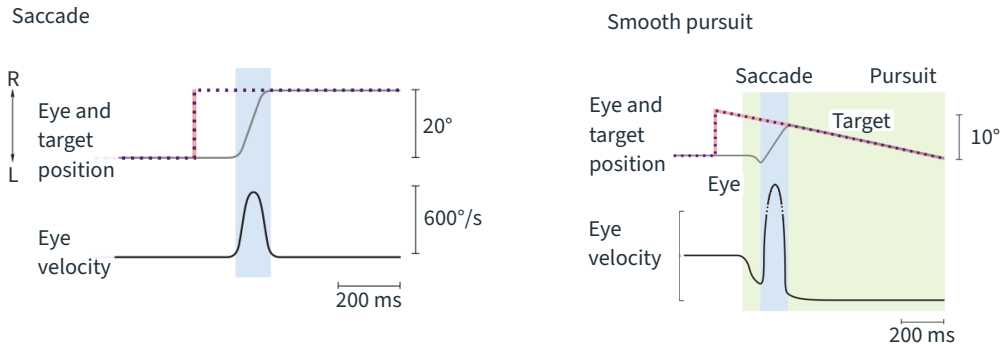
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LECTURE NOTE

Movements can be sorted by function or by control mode, as in L01, but the muscles are shared: the breathing muscles act voluntarily, rhythmically and reflexively. The eye is the same: the oculomotor nuclei are the final common targets of every type of eye movement. Whatever system issues the command, it ends on the same motor neurons.

YOUR NOTES

The saccade system and the smooth pursuit system work together



Kandel 6e, Fig. 35–6, p. 867 · human eye position and velocity; A a saccade, B smooth pursuit with a catch-up saccade; labels re-set from the figure

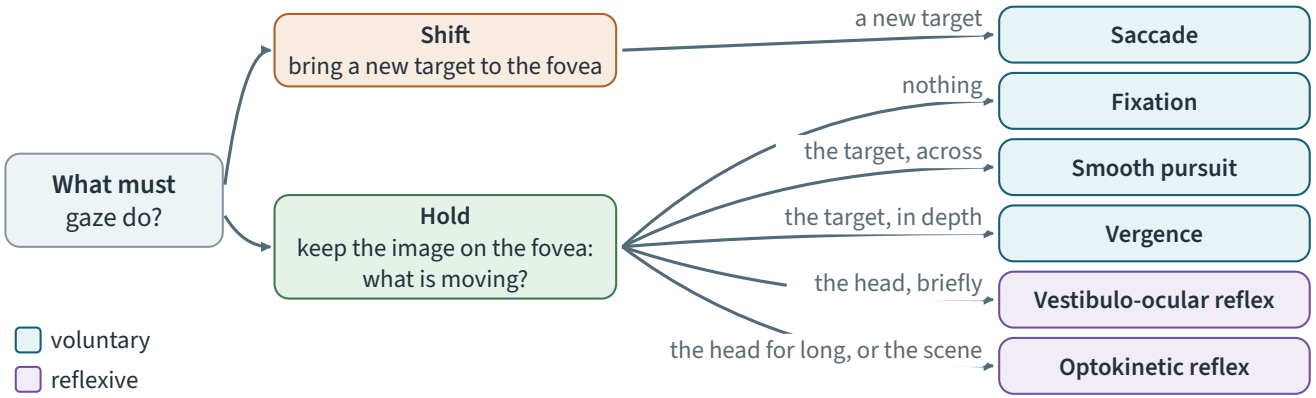
- The saccade:
  - Brief jump with one sharp velocity peak
  - Over before vision can correct it
- Pursuit:
  - Eye velocity matches target velocity
  - Catch-up saccade removes the position error

LECTURE NOTE

A saccade starts about 200 ms after the target jumps and moves the eye with a smooth, stereotyped velocity profile. In pursuit the eye matches the target's velocity, starting before any saccade and briefly moving away from the target; a catch-up saccade then removes the position error. One movement shifts gaze, the other holds it.

YOUR NOTES

Gaze has six systems



Kandel 6e, Ch. 35, p. 866 · the six systems from the text; the shift/hold grouping and the control-mode colours are a teaching synthesis. Body: L01's circuit maps, Kandel 6e, Figs. 33–1, 34–2A, 36–13

- Eye: six systems, one per disturbance.
- Body, as in L01: posture, locomotion, reach and grasp each have their own circuits.

LECTURE NOTE

The eye has six systems: saccades shift gaze; fixation, pursuit, vergence, the vestibulo-ocular and optokinetic reflexes hold it, each against a different disturbance. L01 found the same for the body: posture, locomotion and reaching have differently organized circuits, with the three control modes cutting across them.

YOUR NOTES

Body and eye systems partly overlap

Body

- Spinal interneurons of the locomotor rhythm:
  - Gate reflexes: the same stimulus excites flexors in one phase, extensors in another
  - Carry brainstem and cortical commands to the muscles
- Posture joins voluntary action: ankle torque changes before an arm pull

Eye

- Medial vestibular nucleus and prepositus:
  - Integrator for saccades
  - Eye-velocity neurons for pursuit
  - First relay of the vestibulo-ocular reflex
- The same vestibular nucleus neurons respond to head rotation and to scene motion
- FEF: saccade cells and a pursuit region
- Collicular fixation zone excites omnipause neurons, which gate saccades

> - Body: reflex, rhythmic and voluntary control share spinal circuits.
> - Eye: the systems share brainstem nuclei and cortical areas.
> - In both, the systems are partly distinct and partly shared.

Kandel 6e, Ch. 33, pp. 798–799, 803–805; Fig. 36–8, p. 893 (L01); Ch. 35, pp. 869–870, 873, 875, 878–879; Kandel 5th ed., Ch. 40, pp. 926–928

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LECTURE NOTE

The systems are not sealed off. In the spinal cord the locomotor rhythm gates reflexes and relays descending commands; posture is recruited before a voluntary pull, as L01 showed. For the eye, one set of brainstem nuclei integrates saccades, carries pursuit velocity and relays the vestibular reflex, and FEF serves both saccades and pursuit.

YOUR NOTES

Body and eye combine feedforward and feedback

Feedforward: no error used during the movement

Saccade	visual error, afterwards
Vestibulo-ocular reflex	image slip, over days

Both feedforward errors are used later, to recalibrate the command.

Feedback: a visual error, used throughout

Fixation	position error on the fovea
Smooth pursuit	retinal slip of the target
Vergence	disparity between the eyes
Optokinetic reflex	retinal slip of the scene

Kandel 6e, Ch. 30, pp. 717–719, 735; Ch. 35, pp. 866–867, 878–880; Kandel 5th ed., Ch. 40, pp. 924, 930 · teaching synthesis

- Eye: saccade and vestibulo-ocular reflex feedforward; the holding systems on visual feedback.
- Body, as in L01's reach and grip: feedforward first, feedback after.

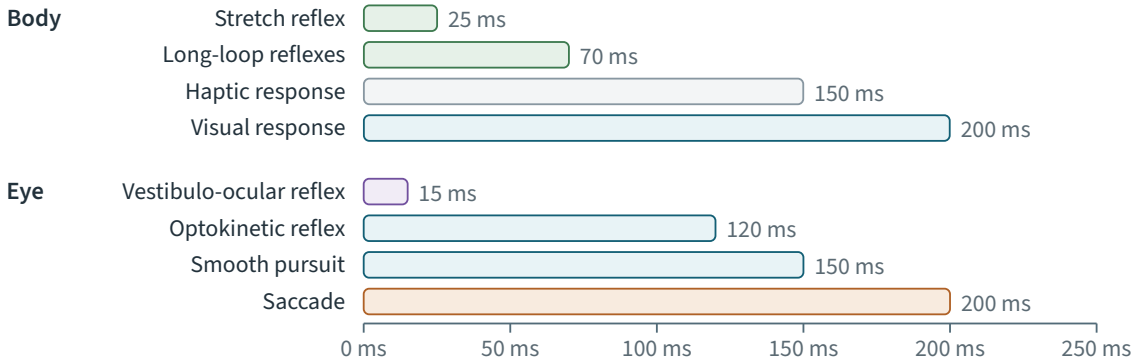
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LECTURE NOTE

Feedforward means the command is issued without waiting for feedback; feedback compares the sensed state with the goal, as in L01. Saccades and the vestibulo-ocular reflex are feedforward, and their errors are used afterwards to recalibrate. Pursuit, fixation, vergence and the optokinetic reflex use visual error throughout. L01's reach and grip examples mixed both.

YOUR NOTES

Multiple feedback loops control body and gaze



Response latencies: instructor, 2026-11-08. Saccade duration: Kandel 6e, Ch. 30, p. 717

- Body: reflexes in 25–70 ms; haptic and visual responses in 150–200 ms.
- Eye:
  - Vestibulo-ocular reflex: 15 ms, the fastest
  - Optokinetic reflex, pursuit, saccade: 120–200 ms, visual delays
  - A saccade lasts ≈30 ms, too brief to steer

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LECTURE NOTE

Reflexes are fast: 25 ms for the stretch reflex, 70 ms for long-loop reflexes, haptic responses take 150 ms and visual responses 200 ms. The eye matches: the vestibulo-ocular reflex responds in 15 ms, the optokinetic reflex in 120 ms, pursuit in 150 ms, a saccade in 200 ms. A saccade lasts about 30 ms, too brief to steer.

YOUR NOTES

Section B summary: gaze is organized like body movement

	Somatic	Eye
Final common path	motor neurons and muscles	oculomotor nuclei and muscles
Systems	posture, locomotion, reach, grasp	six: one shift, five holds
Overlap	shared spinal interneurons	shared brainstem nuclei, FEF
Control	feedforward and feedback	feedforward and feedback
Response latency	25 ms (reflex) to 200 ms (visual)	15 ms (VOR) to 200 ms (saccade)

TAKE HOME

- One final common path; different, partly overlapping systems.
- Feedforward and feedback, on loops with different latencies.

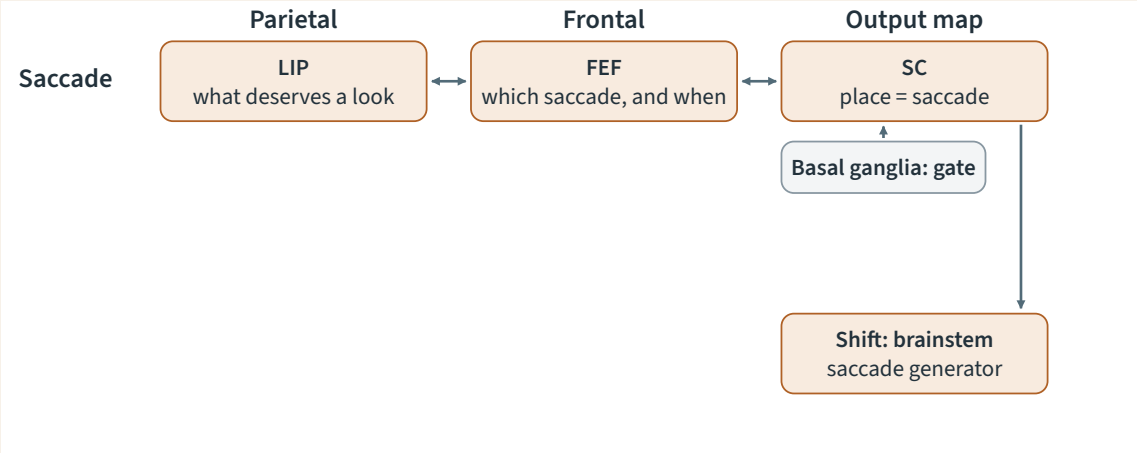
Kandel 6e, Chs. 30, 33 and 35; response latencies: instructor, 2026-11-08 · the section B table, returning

LECTURE NOTE

The section B table returns. Gaze control is organized like control of the body: many systems end on one final common path, they overlap in part, and each mixes feedforward and feedback on loops with different latencies. Within that plan, choosing where to look resembles choosing where to reach.

YOUR NOTES

C · Saccade target selection



Kandel 6e, Ch. 35 · lecture organizer, built up section by section; a teaching schematic, not measured wiring

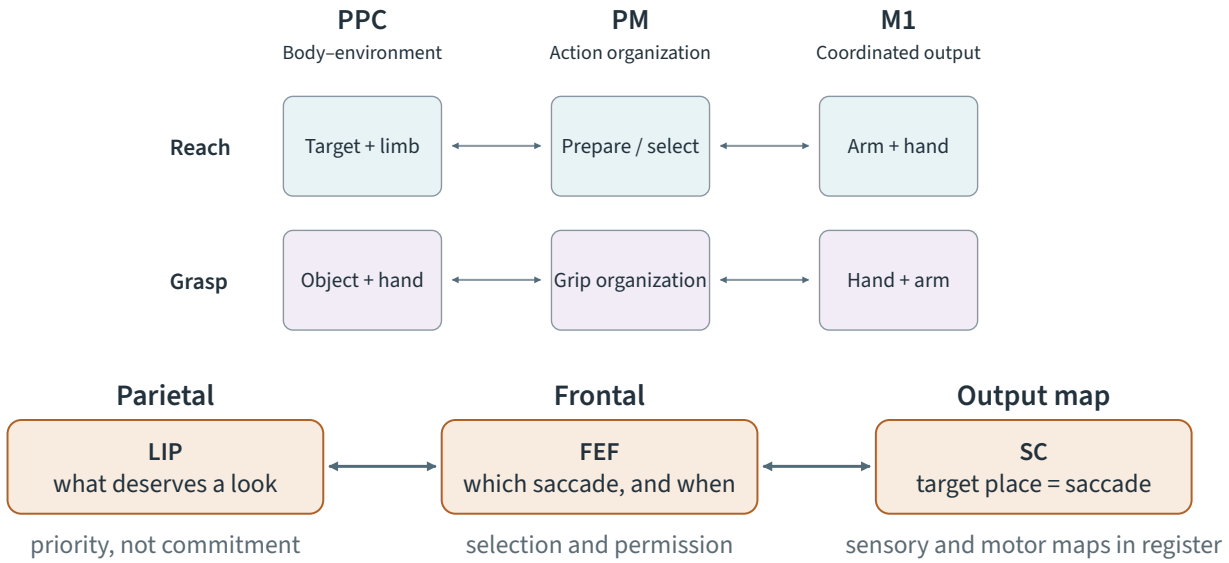
- Parietal and frontal cortex pick the target.
- The colliculus turns its place into a saccade, gated by the basal ganglia.

LECTURE NOTE

The map of the gaze system, built up section by section. Here only the choice of a saccade: LIP marks what deserves a look, FEF picks the saccade and its timing, and the colliculus holds a map in which a place is a movement, gated by the basal ganglia. Its output goes to a brainstem generator, the next section.

YOUR NOTES

Saccades and reaches share three stages



Kandel 6e, Chs. 34 and 35 · two teaching schematics in the same form: PPC / premotor / M1 above, LIP / FEF / SC below

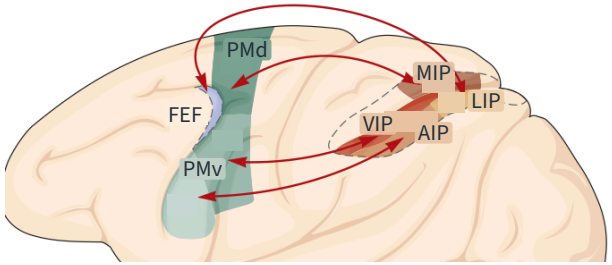
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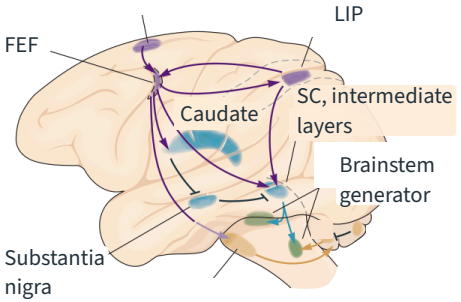
LECTURE NOTE

The cortical voluntary system was organized as parietal, premotor and primary motor stages. The saccade system has the same three columns: LIP marks what deserves a look, FEF selects the saccade and its timing, and the superior colliculus holds the output map. The parallel stops there: the colliculus is a sensory-to-motor map, not a motor cortex.

YOUR NOTES

LIP and FEF form a parietal–frontal pair





Kandel 5th ed., Fig. 29–14, p. 650 · intraparietal areas and their frontal partners; Kandel 6e, Fig. 35–9A, p. 871 · saccade pathways in the monkey; labels re-set from the figures

- Parietal → frontal pairs:
  - LIP → FEF: gaze
  - MIP → PMd: reach
  - AIP → PMv: grasp
- LIP and FEF both project to the colliculus, which drives the brainstem.
- The basal ganglia sit on the way: caudate, then substantia nigra.

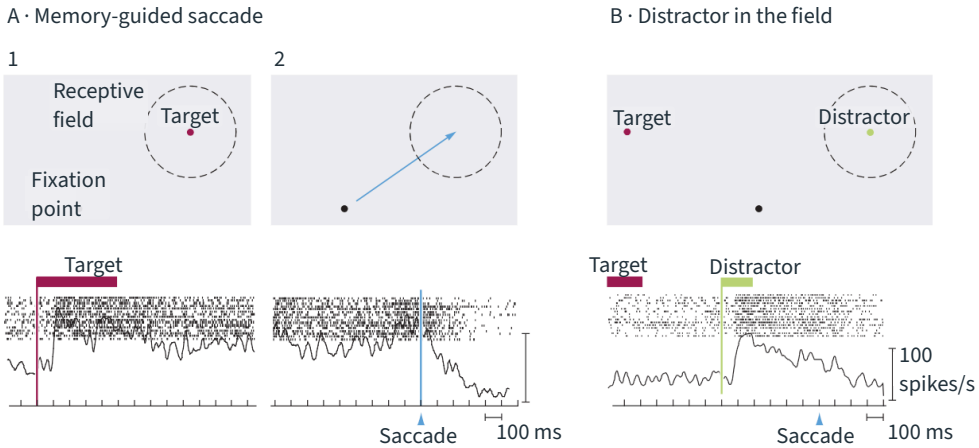
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LECTURE NOTE

Anatomically the saccade pair is one of the parietal–frontal pairs: LIP to the frontal eye field, beside MIP to dorsal premotor cortex for reaching and AIP to ventral premotor cortex for grasping. The pathway figure adds the superior colliculus, the caudate and substantia nigra, and the brainstem generator.

YOUR NOTES

LIP neurons fire for targets and distractors



Kandel 6e, Fig. 35–12, p. 874 · single LIP neuron, memory-guided saccade task (Powell & Goldberg 2000); panel headings replaced, they state the result

- The cell fires from target to saccade, and as much for a distractor.
- So LIP marks **priority**, not a committed plan.

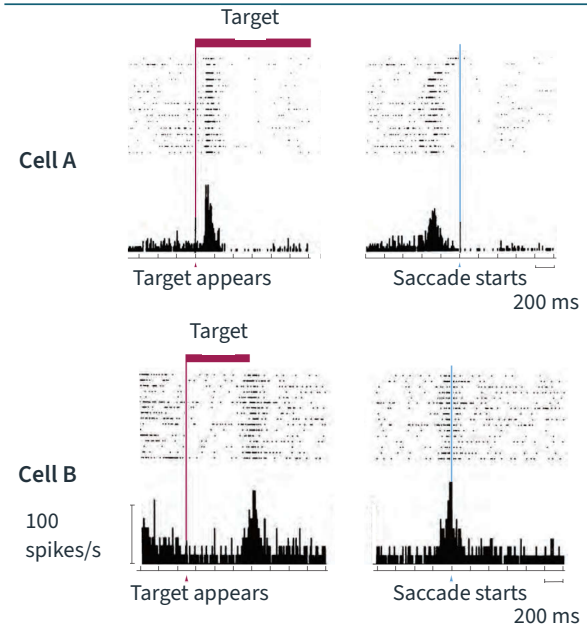
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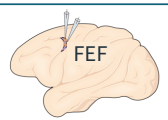
LECTURE NOTE

An LIP neuron fires from the target's appearance until the saccade, holding the remembered location, but it responds just as strongly to a distractor flashed in its field while a saccade is planned elsewhere. Its activity marks what deserves attention rather than a committed movement. In A the monkey looks to the remembered place; in B it looks elsewhere.

YOUR NOTES

Aligning on two events shows what an FEF cell encodes





- Stimulating FEF evokes saccades

- Cell A follows the target: a visual cell.
- Cell B follows the saccade: a movement cell.

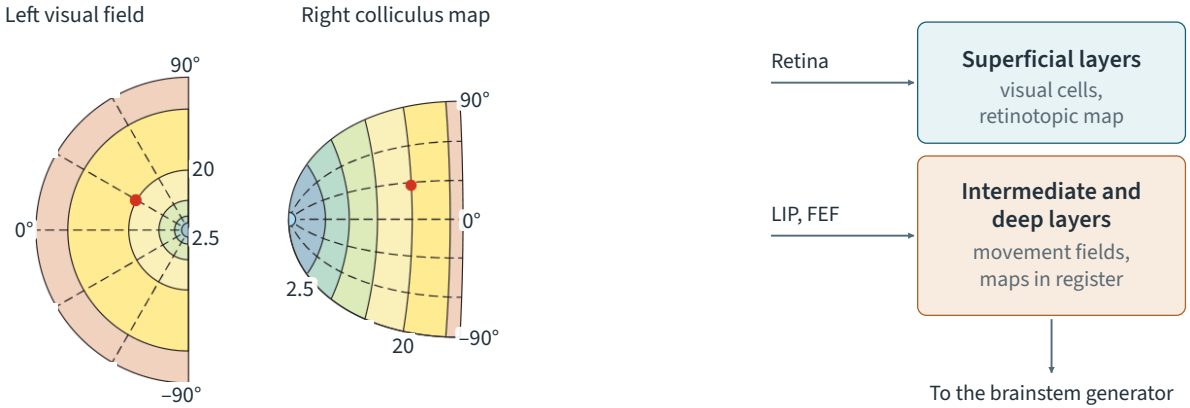
Kandel 6e, Fig. 35–13, p. 876 · two FEF neurons (Bruce & Goldberg 1985); panel headings removed, they state the answer; labels re-set from the figure

LECTURE NOTE

Re-aligning the same spikes on target onset and on saccade onset shows which event a cell is tied to. One cell's burst is sharp at the target and smeared at the saccade; the other is the reverse. FEF movement cells fire only before relevant saccades; frontal lesions impair anti-saccades and memory-guided saccades.

YOUR NOTES

The colliculus maps the visual field



Kandel 6e, Fig. 35–10, p. 872 · visual field and collicular map in polar coordinates (Quaia et al. 1998), a subset of labels re-set; layers: course schematic after pp. 871–873

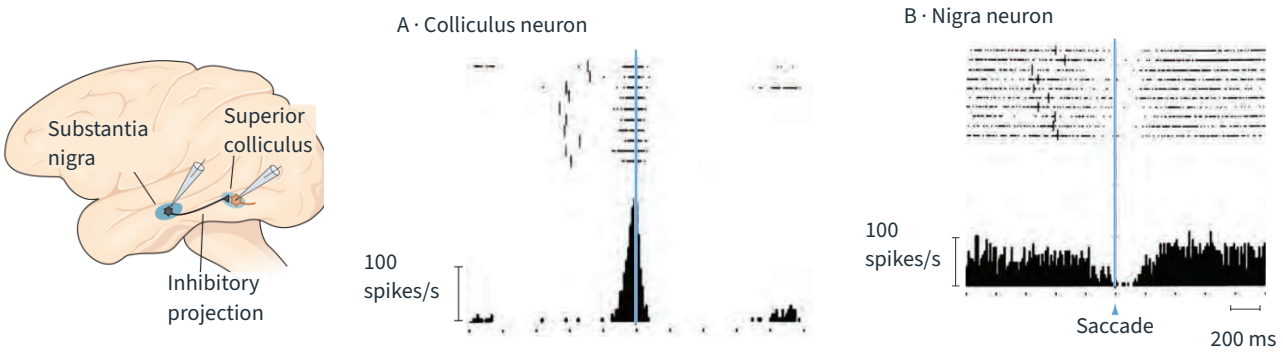
- Each place in the field has its own place on the map.
- Visual and movement maps lie in register: a place on the map **is** a saccade vector.
- A reach must relate target and hand; a saccade needs only the place.

LECTURE NOTE

The superficial layers hold a visual map; the intermediate and deep layers hold movement fields. The visual layers reach the movement layers mainly by a loop through the thalamus (pulvinar) and cortex. Stimulating a site evokes a saccade to the matching place, so a target's place specifies the saccade vector. The population of broad movement fields sets it.

YOUR NOTES

The basal ganglia gate the colliculus



Kandel 6e, Fig. 35–11, p. 873 · superior colliculus and substantia nigra (pars reticulata) neurons around a saccade (Hikosaka & Wurtz); panels set side by side, labels re-set

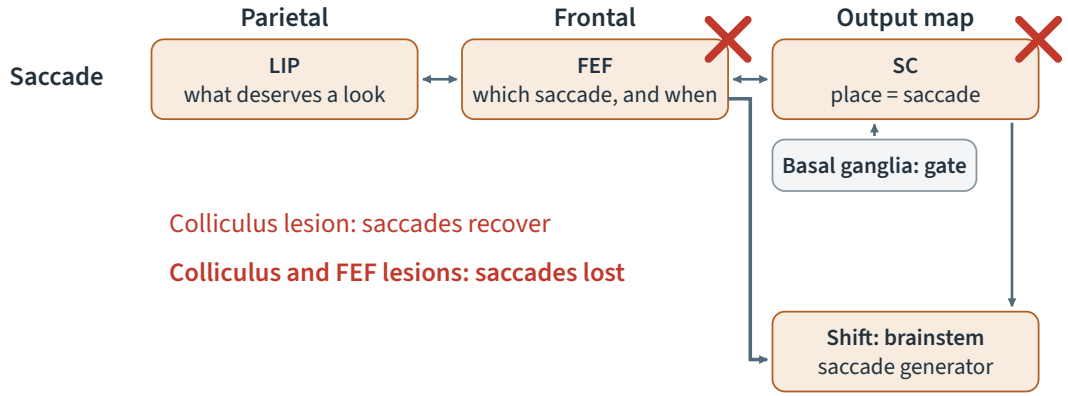
- The nigra pauses as the colliculus bursts: the basal ganglia gate it by disinhibition.

LECTURE NOTE

The nigral neuron fires steadily and falls silent just before the saccade while the collicular neuron bursts: the basal ganglia gate the colliculus by disinhibition, caudate to nigra to colliculus.

YOUR NOTES

FEF reaches the brainstem without the colliculus



Kandel 6e, Ch. 35, pp. 873, 875 · lecture organizer with FEF's own projection to the brainstem reticular formation; lesion outcomes in monkeys

- FEF's route alone is enough, in time.
- Saccades need either the colliculus or FEF.

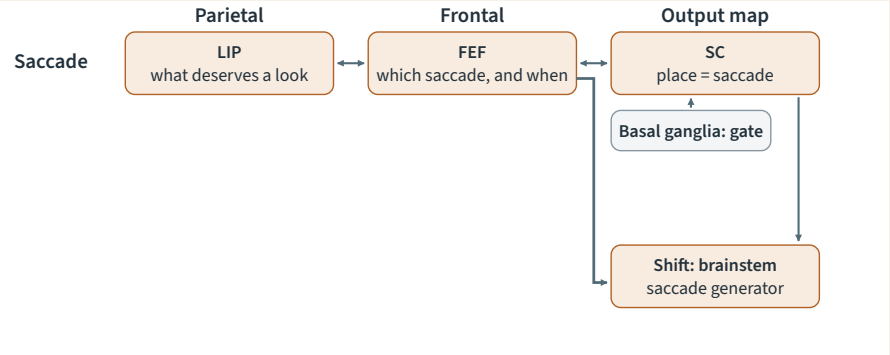
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LECTURE NOTE

FEF projects to the colliculus and also, on its own, to the pontine and mesencephalic reticular formation, though not directly to the burst neurons. After a collicular lesion saccades return because that route remains; only when FEF is also damaged are saccades permanently compromised. Two parallel routes from cortex reach the brainstem.

YOUR NOTES

Section C summary: cortex picks the target, the colliculus moves the eye



TAKE HOME

- LIP marks what deserves a look; FEF picks the saccade.
- The colliculus turns a place into a movement, gated by the basal ganglia.
- FEF also has its own route to the brainstem.

Kandel 6e, Ch. 35, pp. 871–876 · the section C map, returning, with FEF's own route

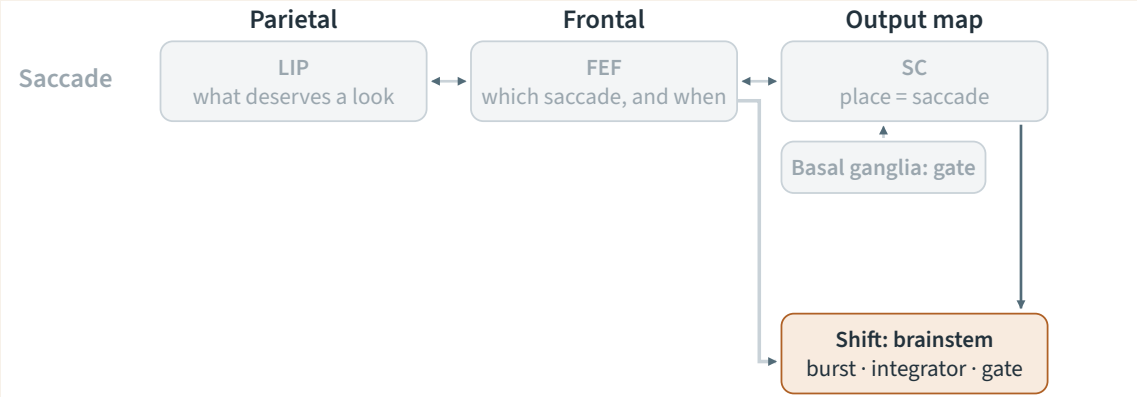
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LECTURE NOTE

The section C map returns. Choosing a saccade follows the parietal, frontal, output-map order of a limb movement: LIP marks priority, FEF selects and times the saccade, and the colliculus turns a retinal place into a movement, gated by the basal ganglia. FEF's own route to the brainstem lets saccades survive a collicular lesion.

YOUR NOTES

D · Saccade generation



Shared integrator and eye motor nuclei: the final common path

Kandel 6e, Ch. 35 · lecture organizer, built up section by section; a teaching schematic, not measured wiring

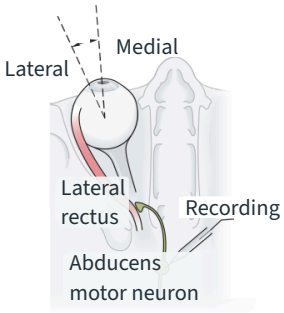
- The brainstem turns the chosen saccade into a burst and a step, on a final common path.

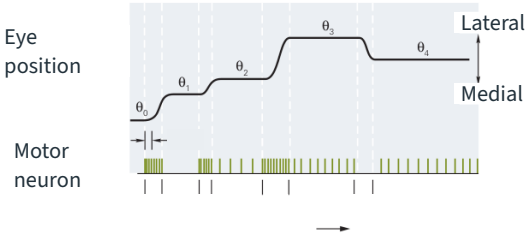
LECTURE NOTE

The same map, now with the brainstem generator in focus. The colliculus hands over a saccade vector; burst neurons, an integrator and omnipause neurons turn it into the motor neurons' pulse and step. The integrator and eye motor nuclei are shared by every eye movement: the final common path.

YOUR NOTES

Recording one abducens motor neuron





- The orbit's load needs a burst against viscosity and a step against elasticity.
- The neuron fires:
  - Burst with each saccade: the pulse
  - Higher rate at each lateral position: the step

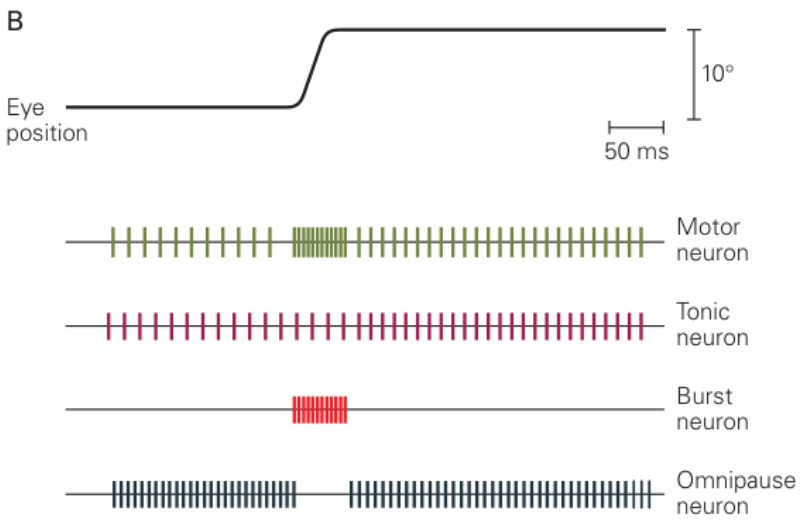
Kandel 6e, Fig. 35–7A, p. 868 · abducens motor neuron in the monkey (Fuchs & Luschei 1970); labels re-set from the figure

LECTURE NOTE

Recording from an abducens motor neuron while the monkey makes saccades toward the side of the lateral rectus shows two components: a burst during each saccade, and a tonic rate that steps up with every more lateral position. The burst is the velocity signal, the tonic rate the position signal; the motor neuron sums them.

YOUR NOTES

Taking the motor neuron’s signal apart, cell by cell



- **Motor neuron**: burst, then a higher steady rate
- **Tonic**: steady rate only, the position
- **Burst**: during the saccade only, the velocity
- **Omnipause**: silent only during the saccade, a gate

Kandel 6e, Fig. 35–8B, p. 869 · firing patterns of four brainstem cell types around a horizontal saccade

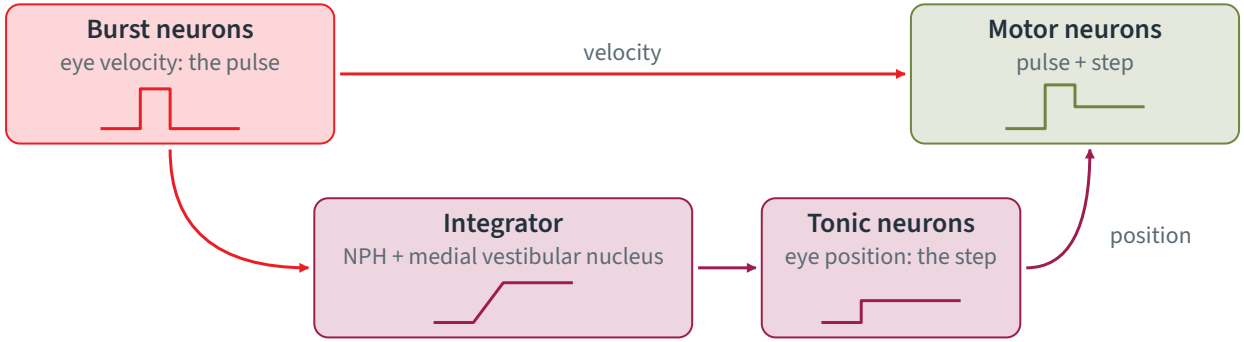
D.03 26 / 44

LECTURE NOTE

The firing patterns separate the motor neuron's signal into cells that carry only position, the tonic neurons, and cells that carry only velocity, the burst neurons. A fourth type, the omnipause neuron, fires continuously except around every saccade; stimulating it arrests a saccade in flight, so it gates the burst.

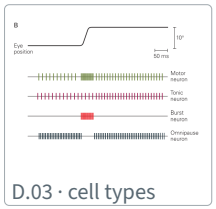
YOUR NOTES

Position is the integral of velocity



Kandel 6e, Fig. 35–8 and p. 870 · teaching simplification: the velocity signal, its integral, and their sum

- The burst drives the motor neurons and, through the integrator, sets the step.
- Without the integrator, the eye drifts back.



D.03 · cell types

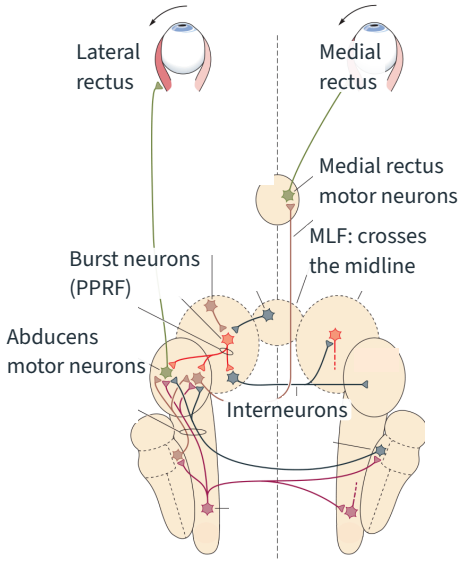
D.04 27 / 44

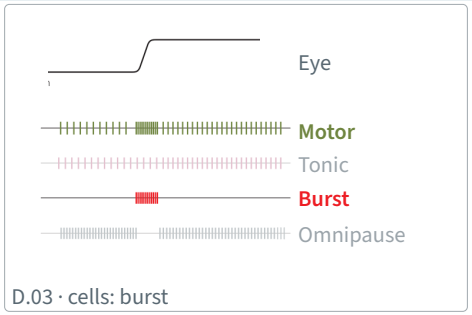
LECTURE NOTE

The step does not need a separate command: integrating the velocity burst over time gives the change in position. The integrator for horizontal movements is the nucleus prepositus hypoglossi and medial vestibular nucleus, with the flocculus. Animals with lesions there make normal saccades but the eyes drift back to centre afterward.

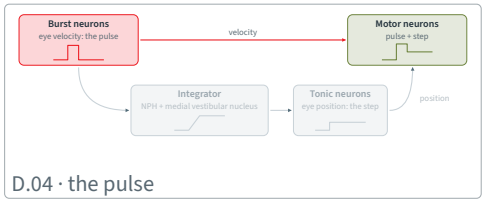
YOUR NOTES

Burst neurons make the pulse





D.03 · cells: burst



D.04 · the pulse

- Burst neurons fire the abducens motor neurons: the pulse.
- Interneurons cross to the other eye: one side moves both eyes.

Kandel 6e, Fig. 35–8A, p. 869 and pp. 869–870 · brainstem circuit for a horizontal saccade to the left; labels re-set, others removed

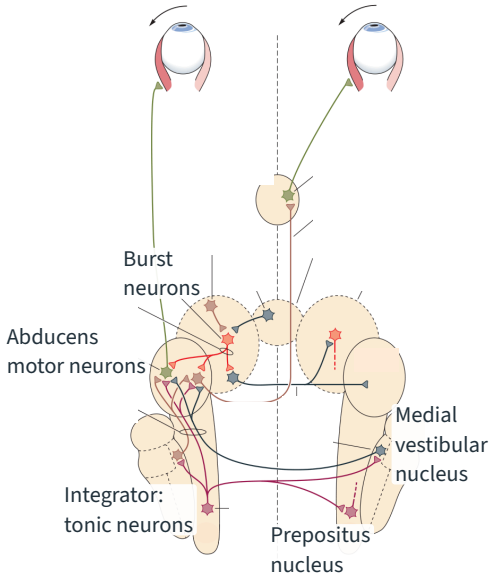
D.05 28 / 44

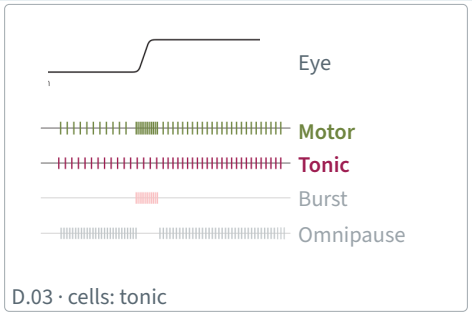
LECTURE NOTE

Excitatory burst neurons in the paramedian pontine reticular formation fire the abducens motor neurons to the lateral rectus: the pulse. Abducens interneurons send the same command across the midline, up the medial longitudinal fasciculus, to the other eye's medial rectus motor neurons. One side of the brainstem moves both eyes.

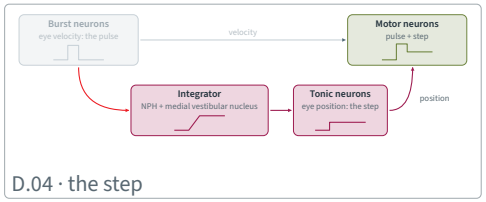
YOUR NOTES

An integrator makes the step





D.03 · cells: tonic



D.04 · the step

- Tonic neurons integrate the burst into a steady rate: the step.
- They lie in the medial vestibular and prepositus nuclei.

Kandel 6e, Fig. 35–8A, p. 869 and pp. 869–870 · brainstem circuit for a horizontal saccade to the left; labels re-set, others removed

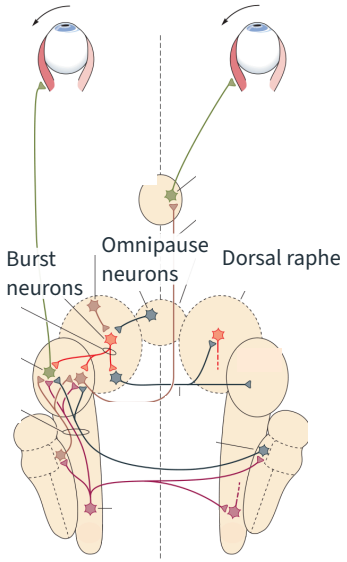
D.06 29 / 44

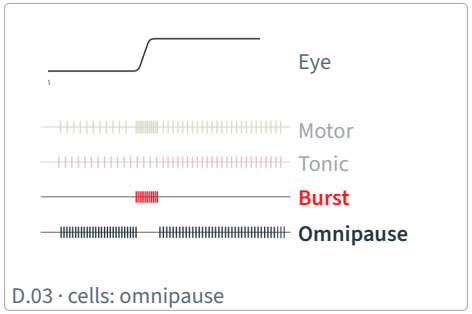
LECTURE NOTE

The burst also reaches tonic neurons that integrate it into a steady rate and pass it to the same motor neurons: the step. They lie in the medial vestibular nucleus and nucleus prepositus hypoglossi, working with the flocculus. Lesions there leave saccades normal, but the eyes drift back afterward.

YOUR NOTES

Omnipause neurons gate the burst





D.03 · cells: omnipause

- Omnipause neurons inhibit the burst neurons between saccades.
- They fall silent before every saccade: the gate.

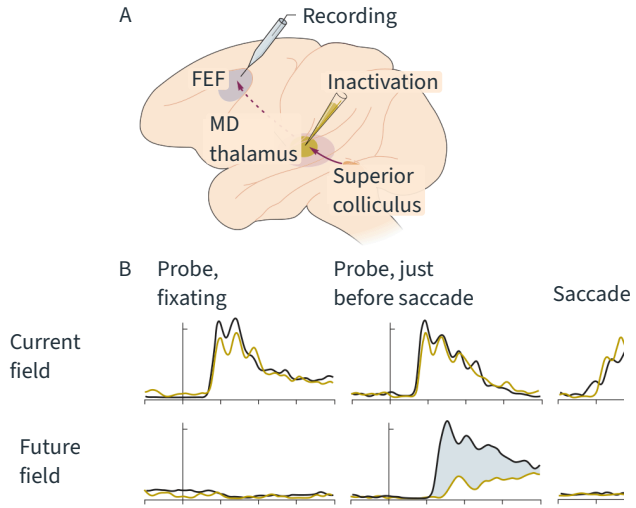
Kandel 6e, Fig. 35–8A, p. 869 and pp. 869–870 · brainstem circuit for a horizontal saccade to the left; labels re-set, others removed

LECTURE NOTE

Omnipause neurons in the dorsal raphe fire continuously and inhibit the burst neurons. They stop shortly before and during every saccade, horizontal or vertical, letting the burst through; stimulating them arrests a saccade in flight. A saccade needs burst neurons excited and omnipause neurons silenced at once.

YOUR NOTES

FEF receives a copy of the saccade command



- A copy of the collicular command reaches FEF through the thalamus.
- Blocking the relay spares the current field but cuts the shift to the future field.

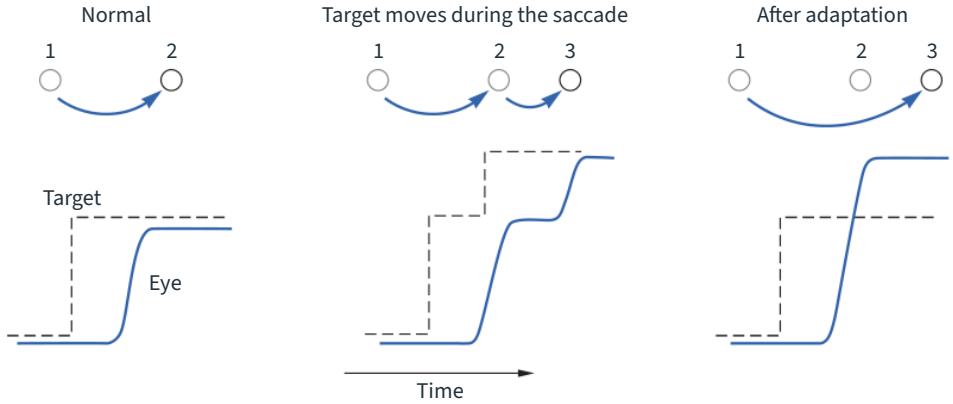
Kandel 5th ed., Fig. 29–11, p. 647 · corollary discharge from colliculus through thalamus to FEF (Sommer & Wurtz); Kandel 6e, Ch. 35, p. 877; labels re-set from the figure

LECTURE NOTE

In B one FEF cell is probed in its current field and where its field will be after the saccade, before and with the thalamic relay inactivated. The command copy, not muscle sense, shifts the field before the saccade.

YOUR NOTES

Saccades recalibrate from their own errors



Kandel 6e, Fig. 37–13B, p. 927; Ch. 37, p. 925; Ch. 35, p. 877 · saccadic adaptation to a target moved during the saccade; labels re-set

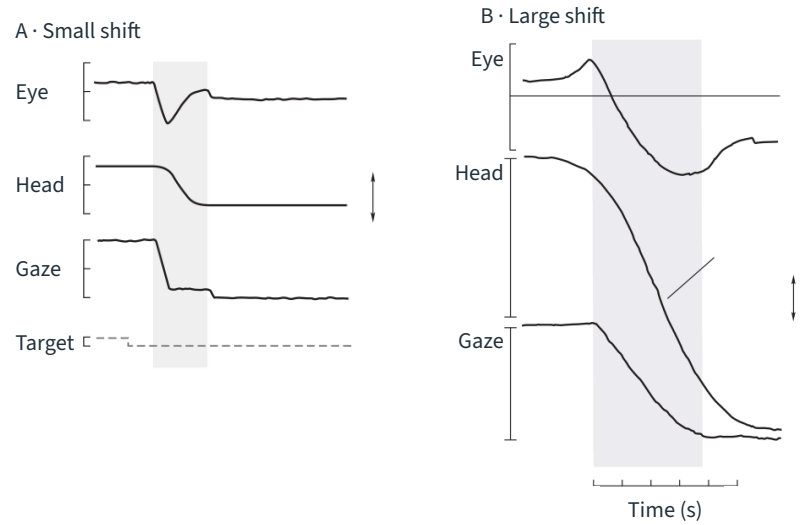
- A target moved mid-saccade leaves an error after the eye lands.
- Over hundreds of trials the first saccade grows: the cerebellum uses the error to recalibrate the next one.

LECTURE NOTE

The error arrives only after the saccade has ended, too late to steer it, so it recalibrates the next one. Climbing fibres to the oculomotor vermis signal the error during adaptation. The same repair keeps saccades accurate when a muscle weakens: patients regain accuracy within days, depending on the vermis and fastigial nucleus.

YOUR NOTES

A gaze shift is a saccade plus a head turn



- Small shift: the saccade first, then the head; the reflex re-centres the eye.
- Large shift: eye and head together; the reflex is held off until the end.
- Eye and head are coordinated to move gaze.

Kandel 6e, Fig. 35–14 and p. 877 · eye, head and gaze during a small and a large gaze shift; panels set side by side, angle and speed scales omitted, labels re-set

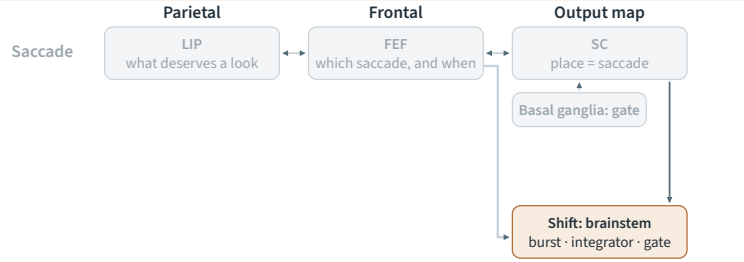
D.10 33 / 44

LECTURE NOTE

A gaze shift is a saccade plus a head turn. In a small shift the saccade lands first; the head follows while the vestibulo-ocular reflex turns the eye back to centre. In a large shift eye and head move together, and the reflex is suppressed until gaze arrives.

YOUR NOTES

Section D summary: the saccade command is built for the load



Shared integrator and eye motor nuclei: the final common path

TAKE HOME

- Pulse: burst neurons · Step: integrator · Gate: omnipause neurons
- One side of the brainstem moves both eyes.
- Errors after the saccade recalibrate the next one.
- Large gaze shifts add the head, with the reflex held off.

Kandel 6e, Ch. 35, pp. 868–870, 877; Ch. 37, p. 925 · the section D map, returning

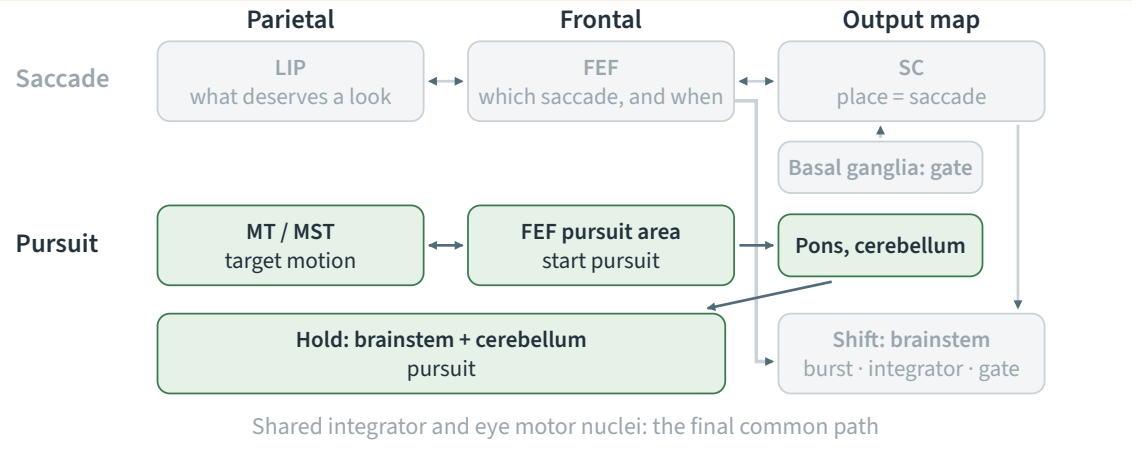
D.11 34 / 44

LECTURE NOTE

The section D map returns. The orbit's load needs a velocity pulse and a position step; burst neurons give the pulse, the integrator derives the step, and omnipause neurons gate both. One side of the brainstem moves both eyes. The error left after a saccade, read by the cerebellum, recalibrates the next. Large gaze shifts add a head turn.

YOUR NOTES

E · Smooth pursuit



Kandel 6e, Ch. 35 · lecture organizer, built up section by section; a teaching schematic, not measured wiring

- Pursuit has its own path: motion areas and FEF, then pons and cerebellum, not the colliculus.

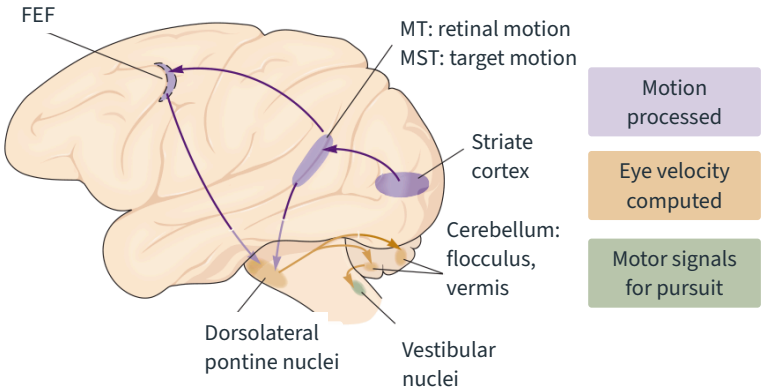
E.01 35 / 44

LECTURE NOTE

The map gains a second row: pursuit. MT and MST signal target motion, the FEF pursuit area starts the movement, and the output runs through the pons and cerebellum to the brainstem holding system, bypassing the colliculus. Pursuit's behaviour, velocity matching with a catch-up saccade, was shown in section B.

YOUR NOTES

Pursuit bypasses the colliculus



Motion processed

Eye velocity computed

Motor signals for pursuit

- MT and MST sense target motion.
- FEF's pursuit area sets the eye velocity.
- Output: pons and cerebellum, not the colliculus.
- Like grasp beside reach, pursuit matches a **property**: velocity.

Kandel 6e, Fig. 35–15, p. 878 · cortical, pontine and cerebellar pathways for smooth pursuit; labels re-set and shortened from the figure

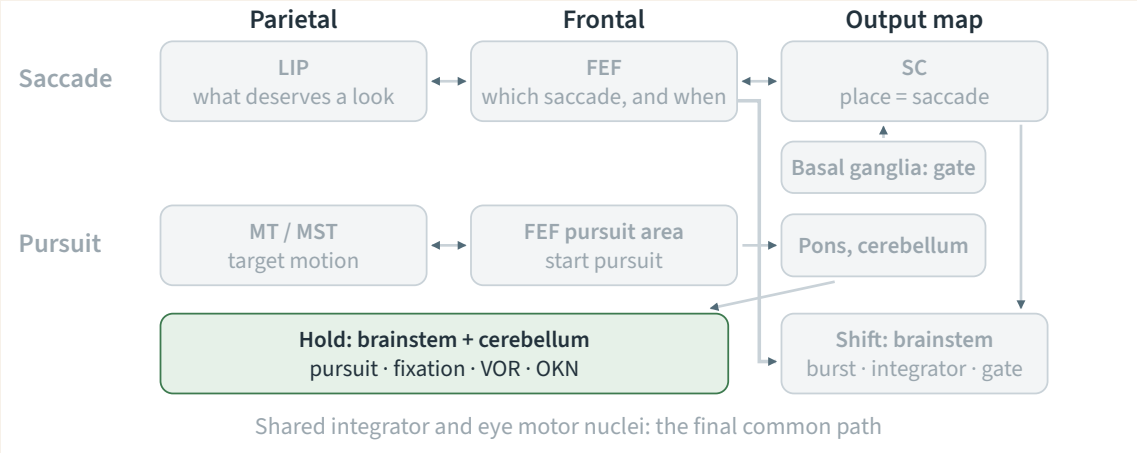
E.02 36 / 44

LECTURE NOTE

Pursuit has its own cortical pair ventral to the saccade areas, MT and MST for target motion and a pursuit region of the frontal eye field, which parallels grasp beside reach: both match a property of the object rather than its place. Its output bypasses the colliculus, running through the pons and cerebellum to the vestibular nuclei.

YOUR NOTES

F · Fixation



Shared integrator and eye motor nuclei: the final common path

Kandel 6e, Ch. 35 · lecture organizer, built up section by section; a teaching schematic, not measured wiring

- Holding gaze: fixation and the vestibular and optokinetic reflexes, on the same brainstem and cerebellum.

LECTURE NOTE

The finished map. The holding systems join pursuit in the brainstem and cerebellum: fixation and the vestibulo-ocular and optokinetic reflexes. All of them, and saccades, share the integrator and the eye motor nuclei.

YOUR NOTES

Fixation is active

- Saccades are suppressed by:
  - Omnipause neurons
  - Rostral colliculus, the fixation zone
- Lesions: distractors capture the eyes

Fixation is not stillness

- Small eye movements continue
  - Without them a still image “rapidly fades to a field of gray”

> - Holding gaze on a still target is a job in its own right.
> - The other holding systems take over when the target or the head moves.

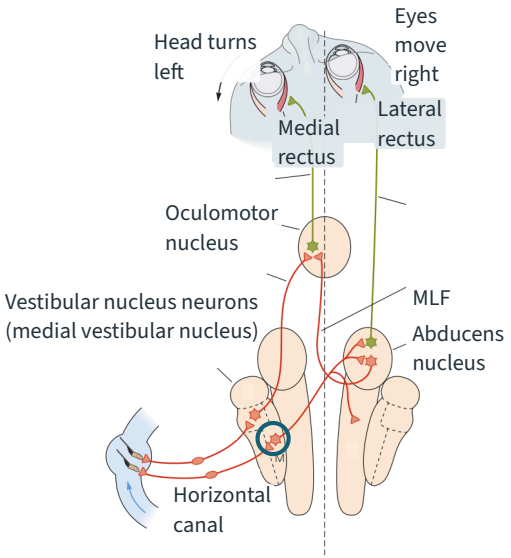
Kandel 6e, Ch. 35, pp. 869, 873; Kandel 5th ed., Ch. 29, p. 637

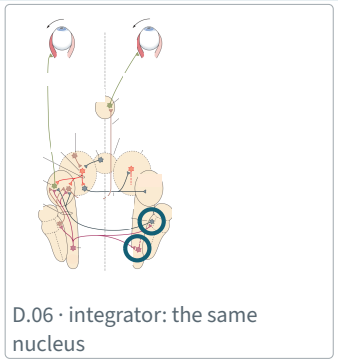
LECTURE NOTE

Fixation is an active system: omnipause neurons and the rostral colliculus suppress saccades, and lesions there let distractors capture gaze. Nor is it stillness; small movements continue, and without them a stationary image fades.

YOUR NOTES

A head turn drives the eyes the opposite way





D.06 · integrator: the same nucleus

- The canal senses head velocity.
- Vestibular nucleus neurons relay it: the saccade integrator's nucleus.
- Abducens, and across the MLF oculomotor, neurons turn both eyes against the head.
- Three neurons, 15 ms, no vision: open loop.

Kandel 5th ed., Fig. 40–11A, p. 927 and p. 926 · excitatory connections of the horizontal reflex, head turning left; Kandel 6e, pp. 869–870 (integrator); labels re-set. Reflex latency: instructor, 2026-11-08

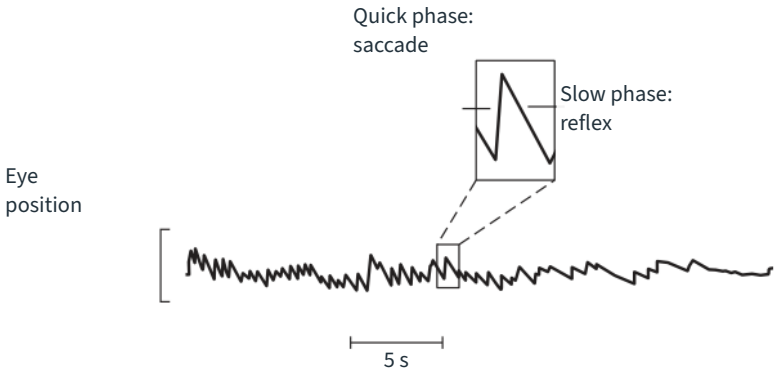
F.03 39 / 44

LECTURE NOTE

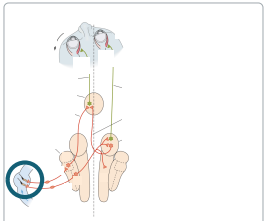
The left horizontal canal senses the head turning left. Vestibular nucleus neurons in the medial vestibular nucleus, the nucleus that also integrates saccades, cross to the right abducens nucleus; its motor neurons pull the right lateral rectus and its interneurons cross back in the MLF to the left medial rectus. Opposite muscles are inhibited. Three neurons, open loop.

YOUR NOTES

Rotation in the dark drives nystagmus



- The slow phase counter-rotates the eye; a quick phase resets it.
- In steady rotation in the dark the canal signal fades, and the nystagmus dies out.



F.03 · the canal

Kandel 5th ed., Fig. 40–8, p. 924 · eye position during constant counterclockwise rotation in the dark (Leigh & Zee 1991); labels re-set

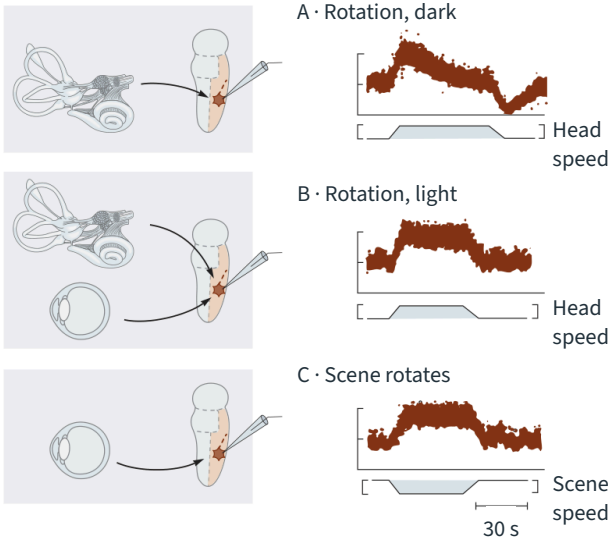
F.04 40 / 44

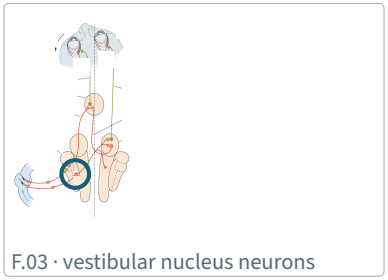
LECTURE NOTE

During constant rotation in the dark the eye drifts against the rotation in slow phases and snaps back in quick saccadic phases. The slow phase gradually slows and stops: the cupula relaxes and the canal signal returns to baseline. Velocity storage in the brainstem prolongs it, but the reflex alone cannot sustain it.

YOUR NOTES

Vision keeps the vestibular nucleus neurons firing





F.03 · vestibular nucleus neurons

- In the dark, firing fades during steady rotation.
- In the light it is sustained, and the scene alone drives it.
- It cannot tell self-motion from scene motion.

Kandel 5th ed., Fig. 40–12, p. 928 and p. 926 · one neuron in the medial vestibular nucleus (Waespe & Henn 1977); firing-rate scales omitted, labels re-set

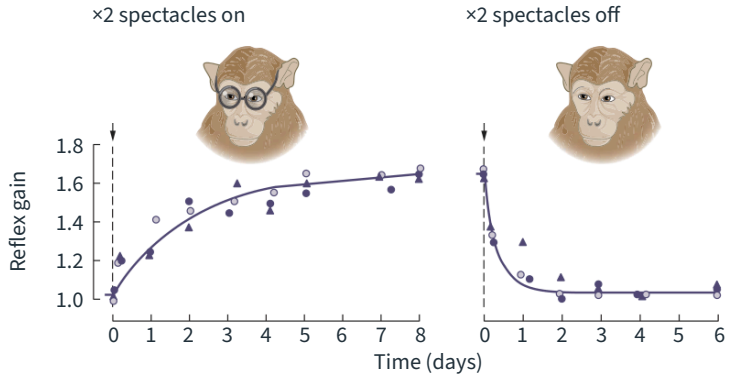
F.05 41 / 44

LECTURE NOTE

One vestibular nucleus neuron, from the arc. In the dark its firing follows the canal signal and fades. In the light the moving scene's optokinetic signal reaches the same neuron and sustains it; moving the scene alone drives it just as well. Vision backs up the fast reflex, and the neuron cannot tell the two apart.

YOUR NOTES

Magnifying spectacles raise the reflex gain



Kandel 5th ed., Fig. 40–14A, p. 930 · reflex gain in a monkey wearing ×2 spectacles, then after removing them (Miles & Eighmy 1980); exposure legend omitted, labels re-set

- Gain climbs toward 2 over days, then falls back.
- The image slip recalibrates the reflex, in the cerebellum.

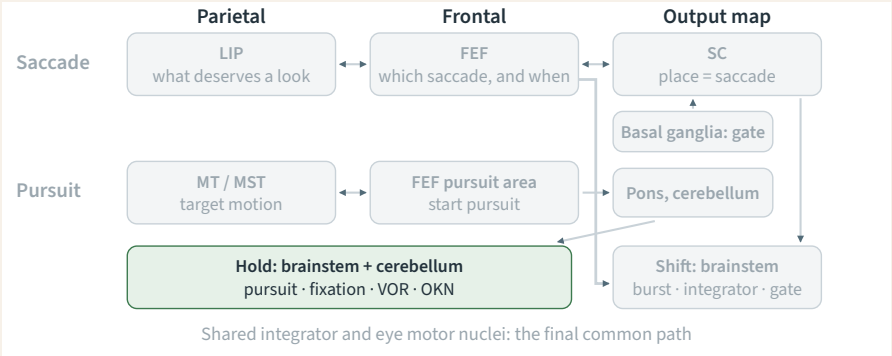
F.06 42 / 44

LECTURE NOTE

An open-loop reflex still needs calibrating. With magnifying spectacles each head turn leaves the image slipping; over days the reflex gain rises to compensate, and returns when the spectacles come off. The retinal slip is the teaching signal and the cerebellum the site of the change.

YOUR NOTES

Section F summary: vision backs up and recalibrates the reflexes



Shared integrator and eye motor nuclei: the final common path

TAKE HOME

- A fast open-loop reflex, backed by slow visual ones, on a shared integrator.
- Vision recalibrates the reflex, through the cerebellum.

Kandel 6e, Ch. 35; Kandel 5th ed., Ch. 40 · the finished map, returning

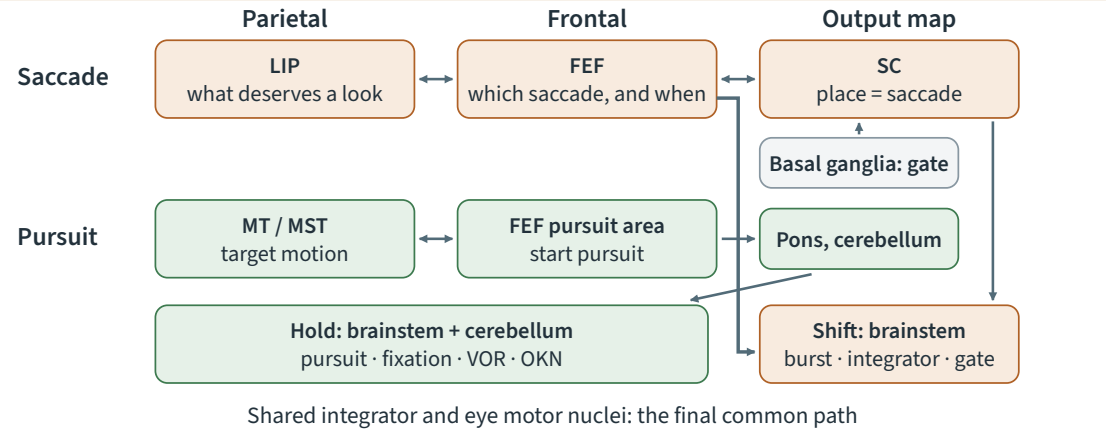
F.07 43 / 44

LECTURE NOTE

The finished map returns. Holding gaze combines an open-loop vestibular reflex that is fast but fades with closed-loop visual reflexes that are slow but sustained, all on the same neurons and integrator as saccades and pursuit, and a reflex gain recalibrated by its own visual error through the cerebellum.

YOUR NOTES

Lecture summary: a simple effector, controlled like the body



Shared integrator and eye motor nuclei: the final common path

Kandel 6e, Chs. 30, 35, 37; Kandel 5th ed., Chs. 29, 40 · the lecture organizer, complete

TAKE HOME

- The eye is a simple, predictable load: commands need not wait for feedback.
- Gaze is organized like the body: overlapping systems, one final common path.
- Shifting gaze: cortex picks the target, the colliculus turns place into movement, the brainstem builds it.
- Holding gaze: reflexes and pursuit share the integrator, and vision recalibrates them.

LECTURE NOTE

The whole map, every part live. The eye is a simple load, so its commands can be planned. Its control is organized like the body's. Cortex and colliculus choose a saccade and the brainstem builds it; the holding systems keep gaze still on the same integrator and motor nuclei, and their errors recalibrate them through the cerebellum.

YOUR NOTES